
Time Series Analysis for Quantitative Finance



INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH KOLKATA

Project Report
Kalaap Sadhu

Under the guidance of Dr. Anirvan Chakraborty, IISER Kolkata

Part of a series of notes available at kalaapks.github.io/notes.html

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List of Figures

1.1	Minute-to-minute data for 16 days (green) vs daily data for 16 days (red)	2
1.2	Joint distribution of discrete $X - \mu_X$ and $Y - \mu_Y$	3
1.3	$Y_t = A \cos(\theta t) + B \sin(\theta t)$ where $A, B \sim \mathcal{N}(0, 1)$	7
1.4	Trend, Seasonality, Heteroscedasticity	10
1.5	A single path of White noise process	12
1.6	Simulation of 250 paths of a Random Walk process	12
1.7	Autocovariance function of an MA(q) process: it vanishes for all lags $h > q$ (here $q = 6$).	14
1.8	The unit circle (solid), the circle of radius $1 + \varepsilon/2$, and the dashed circle of radius $1 + \varepsilon$; the zeros of ϕ (red dots) all lie outside the dashed circle.	24

Contents

1	Time Series Analysis	1
1.1	Introduction to Time Series data	1
1.1.1	Sampling and Notation	2
1.1.2	Stationarity	5
1.1.3	Time Series Decomposition	11
1.2	Basic models	11

TIME SERIES ANALYSIS

1

As opposed to what we have been seeing in other Statistics papers, $X_1, \dots, X_n$ is no longer going to be i.i.d. $\text{dist}(\theta)$. A classic dataset for forecasting looks like $\{X_t, X_{t+1}, \dots, X_{t+n}\}$ where they are no longer independent and we have the time variable t . Hence, what we are going to do is also called *Time Series Analysis*.

- Classic inference: Data is i.i.d.
- Time Series: Data is ordered in time variable t (not independent).

1.1 Introduction to Time Series data

Steps involved in the analysis of Time Series data:

1. Plot
2. Study the past behaviour
3. Identify underlying pattern
4. Do forecasting / fitting of data (the two statistical problems of time series).

Data Types.

- Cross-sectional data: Observations from different entities at a single point in time. eg. closing stock prices of 20 stocks on a particular day.
- Time series data: Observations from a single entity over different time points. eg. closing stock price of a stock over 6 months.
- Panel / Longitudinal data: Observations from different entities over different time points. eg. closing stock prices of 20 stocks over 6 months.

1.1.1 Sampling and Notation

Notation for Time Series data: $\{Y_t, Y_{t+h}, Y_{t+2h}, \dots\}$ where h is the time between two observations and $\frac{1}{h}$ is the sampling frequency.

Due to modern technology, data can be recorded at very high frequency. So, we must be careful in picking the correct time scale required for the forecast. For example, to forecast the next day's sale at a grocery store, we will use daily aggregates instead of minute-by-minute sales data which is filled with noise.

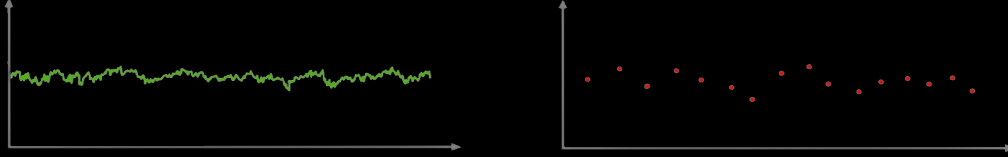


Figure 1.1: Minute-to-minute data for 16 days (green) vs daily data for 16 days (red)

Assume we simply calculate the average to get the next day's sale. Computationally, the minute by minute method is the average of $(60 \times 24 \times 16) = 23040$ numbers, whereas the later is the average of 16 numbers only. Choosing the first time scale is computationally more expensive because there is so much noise.

Let us fix notations now for the entire discussion:

- Time Series: $\{Y_t, Y_{t+h}, Y_{t+2h}, \dots\}$
- Mean: $\mu_t = \mathbb{E}[Y_t]$
- Variance: $\sigma_t^2 = \text{Var}(Y_t) = \mathbb{E}[(Y_t - \mu_t)^2] = \mathbb{E}[Y_t^2] - \mu_t^2$
- Autocovariance: $\gamma_Y(t, s) = \text{Cov}(Y_t, Y_s) = \mathbb{E}[(Y_t - \mu_t)(Y_s - \mu_s)]$
- Autocorrelation: $\rho_Y(t, s) = \frac{\gamma_Y(t, s)}{\sigma_t \sigma_s}$

Remark 1.1. The “auto” in these names comes from the fact that they are covariance and correlation of the same series at two different time points.

In general, the population mean, variance, etc., are unknown and hence $\gamma_Y(t, s)$ cannot be computed directly. Without the knowledge of the precise distribution of the population, we cannot compute the expectation. We will replace the population mean ($\mathbb{E}$) by the sample average.

Let $y_1, \dots, y_n$ be our sample or observed time series, and suppose we are trying to forecast Y_t at $t = n+1$, for example. Let $\bar{y} = \frac{1}{n} \sum_{j=1}^n y_j$.

- $\mathbb{E}[(Y_t - \mu_t)^2]$ will be replaced by $\frac{1}{n} \sum_{j=1}^n (y_j - \bar{y})^2$
- $\mathbb{E}[(Y_t - \mu_t)(Y_{t+k} - \mu_{t+k})]$ will be replaced by $\frac{1}{n} \sum_{j=1}^{n-k} (y_j - \bar{y})(y_{j+k} - \bar{y})$

Definition 1.2. Sample Autocorrelation Function of Lag k – ACF(k) : Let $y_1, \dots, y_n$ be the observed time series (sample of size n) and $\bar{y} = \frac{1}{n} \sum_{j=1}^n y_j$ be the sample mean.

$$\text{ACF}(k) = \frac{\sum_{i=1}^{n-k} (y_i - \bar{y})(y_{i+k} - \bar{y})}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (1.1)$$

Note there is a pattern to these pairings; this is almost like usual inner product but with a offset of k -steps.

Remark 1.3. With n -many observations, the Sample ACF of order k has $n - k$ terms in the numerator since we leave out k -many in the second collection before starting the pair-ups. The sample $\text{ACF}(k)$ is the sample version of $\rho_Y(t, t + k)$.

Geometric Interpretation of What ACF Measures

For that, let us start with the population autocorrelation, since when we look at the sample ACF, the meaning will not change. Removing the time stamps, for random variables X and Y :

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}, \quad \text{where } \text{Cov}(X,Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$$

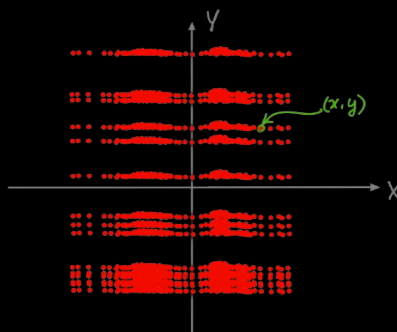


Figure 1.2: Joint distribution of discrete $X - \mu_X$ and $Y - \mu_Y$

$X - \mu_X$ and $Y - \mu_Y$ are centered around 0 individually. Looking at a particular (x, y) joint distribution means looking at an output x of X translated by μ_X , and output y of Y translated by μ_Y . Now, $x - \mu_X > 0$ and $y - \mu_Y > 0$ means x and y lie to the right of the origin on their respective axes; similarly, being < 0 means being to the left of 0 on their respective axes:

- $(x - \mu_X)(y - \mu_Y) > 0 \implies$ both on left or both on right $\implies$ 1st / 3rd quadrant.
- $(x - \mu_X)(y - \mu_Y) < 0 \implies$ one on left and the other on right $\implies$ 2nd / 4th quadrant.

Now, when we vary x and y and take an average, we get the “overall / net” alignment. That is exactly what $\mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \text{Cov}(X, Y)$ is. More precisely, we measure whether both X and Y are in the same direction or different directions; this is exactly what linear alignment is.

$\text{Cov}(X, Y)$ measures if X and Y are related linearly (i.e., if $Y = a + bX$) and ρ_{XY} is the normalized $\text{Cov}(X, Y)$:

$$\rho_{XY} := \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\langle X_c, Y_c \rangle}{\|X_c\| \|Y_c\|} = \langle \hat{X}_c, \hat{Y}_c \rangle$$

which literally means ρ_{XY} measures how the centered X_c and centered Y_c are aligned.

Example 1.4. Suppose $Y = X^2$ (clearly, Y depends completely on X , but not linearly). Let X be symmetric about 0, so odd moments are 0. Then:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0 - 0 = 0$$

This demonstrates that covariance “sees” linear relationships only.

Returning to time series, $\gamma_Y(t, t+k)$ measures linear alignment of centered and normalized versions of random variables Y_t and Y_{t+k} .

Sample ACF Interpretation via Vectors: Consider the vector $\vec{y} = (y_1, y_2, \dots, y_n)$. Define:

$$\begin{aligned}\vec{y}^\ell &= (y_1, y_2, \dots, y_\ell) \quad (\text{the first } \ell \text{ coordinates of } \vec{y}), \\ \vec{y}_\ell &= (y_{\ell+1}, \dots, y_n) \quad (\text{removing the first } \ell \text{ coordinates of } \vec{y}).\end{aligned}$$

Then

$$\begin{aligned}\text{ACF}(1) &= \frac{1}{\|\vec{y} - \bar{y}\|_2^2} \langle \vec{y}^{n-1} - \bar{y}, \vec{y}_1 - \bar{y} \rangle, \\ \text{ACF}(2) &= \frac{1}{\|\vec{y} - \bar{y}\|_2^2} \langle \vec{y}^{n-2} - \bar{y}, \vec{y}_2 - \bar{y} \rangle, \\ \text{ACF}(k) &= \frac{1}{\|\vec{y} - \bar{y}\|_2^2} \langle \vec{y}^{n-k} - \bar{y}, \vec{y}_k - \bar{y} \rangle,\end{aligned}$$

$\text{ACF}(k)$ normalizes the $(n-k)$ -length vectors $\vec{y}^{n-k} - \bar{y}$ and $\vec{y}_k - \bar{y}$ by the full length of $\vec{y} - \bar{y}$ (hence they are strictly inside the unit ball of $\mathbb{R}^{n-k}$), centers them, and checks alignment in the same sense as before.

Definition 1.5. Partial Autocorrelation Function of Order k – $\text{PACF}(k)$: It is the correlation coefficient of Y_t and Y_{t+k} conditional on all intermediate values of the process:

$$\text{PACF}(k) := \text{Corr} \left(Y_t - \mathbb{E}[Y_t \mid Y_{t+1}, \dots, Y_{t+k-1}], Y_{t+k} - \mathbb{E}[Y_{t+k} \mid Y_{t+1}, \dots, Y_{t+k-1}] \right) \quad (1.2)$$

Remark 1.6. Why do we need the $\text{PACF}(k)$ now if $\text{ACF}(k)$ is already there?

$\text{ACF}(k)$ compares $(y_1, \dots, y_{n-k})$ and $(y_{1+k}, \dots, y_n)$. If Y_t influences Y_{t+1} and Y_{t+1} influences Y_{t+2} , then Y_t influences Y_{t+2} . There is no way for ACF to distinguish between direct influence and indirect influence through intermediate lags. So, we have the following plan. Once we project Y_t and Y_{t+k} onto $\mathcal{G}_k = \sigma(1, Y_{t+1}, \dots, Y_{t+k-1})$, whatever remains is the direct influence since all intermediate contamination is gone.

Further elaboration on above Remark: PACF is primarily used in its sample-point version itself, that is, when between Y_t and Y_{t+k} we have $k-1$ steps precisely. There is a continuous-time version of this too but we need not focus much on that since they are not useful in practice.

We know $\mathbb{E}[Y_t \mid \mathcal{G}_k]$ is the projection of Y_t on $\mathcal{G}_k$ in abstract $L^2(\Omega)$ sense, and then $Y_t - \mathbb{E}[Y_t \mid \mathcal{G}_k]$ represents the component of Y_t orthogonal to the information or influence exerted via $Y_{t+1}, \dots, Y_{t+k-1}$. Similarly for $Y_{t+k} - \mathbb{E}[Y_{t+k} \mid \mathcal{G}_k]$.

Now, we measure the linear influence of $Y_t - \mathbb{E}[Y_t \mid \mathcal{G}]$ and $Y_{t+k} - \mathbb{E}[Y_{t+k} \mid \mathcal{G}_k]$ on one another by taking their product and averaging:

$$\text{PACF}(k) = \text{Corr} \left(Y_t - \mathbb{E}[Y_t \mid \mathcal{G}_k], Y_{t+k} - \mathbb{E}[Y_{t+k} \mid \mathcal{G}_k] \right) \quad (1.3)$$

Other Representation: $\mathbb{E}[Y_t \mid \mathcal{G}_k]$ is the linear regression of Y_t using intermediate data:

$$\hat{Y}_t = \mathbb{E}[Y_t \mid \mathcal{G}_k] = \beta_0 + \beta_1 Y_{t+1} + \dots + \beta_{k-1} Y_{t+k-1}$$

So $Y_t - \hat{Y}_t$ is Y_t with the linear dependence on $Y_{t+1}, \dots, Y_{t+k-1}$ removed. We say $\text{PACF}(k)$ is the autocorrelation of Y_t and Y_{t+k} with linear dependence on $Y_{t+1}, \dots, Y_{t+k-1}$ (intermediate steps) removed. We can put $\hat{Y}_t$ in 1.3 and get the explicit formulae.

The structure of continuous time PACF is based on the exact same idea. Here the intermediate space is a continuum and we have to remove all of it. We define,

$$\mathcal{G}_h = \overline{\text{span}}\{Y_{t+s} : 0 < s < h\}$$

Similar to how earlier $\mathcal{G}_k$ did not have Y_t and Y_{t+k} , here too we do not include $s = 0, h$. The continuous-time PACF at lag $h > 0$ is defined as:

$$\text{PACF}(h) = \text{Corr} \left(Y_t - \mathbb{E}[Y_t \mid \mathcal{G}_h], Y_{t+h} - \mathbb{E}[Y_{t+h} \mid \mathcal{G}_h] \right)$$

1.1.2 Stationarity

Let f and F denote the usual probability density function and cumulative distribution function respectively. Note that, due to lack of independence:

$$f_{Y_t, Y_s}(y_t, y_s) \neq f_{Y_t}(y_t)f_{Y_s}(y_s)$$

Handling the joint distribution is hard since $f_{Y_{t_1}, \dots, Y_{t_n}}$ will not factorize. That brings us to the concept of stationarity.

Definition 1.7. Weak Stationarity : First and second-order moments are unaffected by a change of time origin. This is often times called just Stationarity. That is:

1. $\mathbb{E}[Y_t] = \mu < \infty \quad \forall t$
2. $\mathbb{E}[Y_t^2] < \infty \quad \forall t$
3. $\gamma_Y(t, s) = \gamma_Y(t + h, s + h) \quad \forall t, s \in T, h \in \mathbb{Z}$

Remark 1.8. Since both forward and backward translations are allowed, pick $h = -s$. Then:

$$\gamma_Y(t, s) = \gamma_Y(t - s, 0)$$

It is enough to show that $\gamma_Y(t, s)$ is a function of $(t - s)$, since this means the covariance only depends on the gap between t and s , and not on the absolute locations of t and s .

Definition 1.9. Strong Stationarity : For any $\{t_1, \dots, t_k\} \in T$ and $h \in \mathbb{Z}$, the joint distribution of $\{Y_{t_1}, \dots, Y_{t_k}\}$ and $\{Y_{t_1+h}, \dots, Y_{t_k+h}\}$ are identical.

These are stronger conditions on the distribution as opposed to what we saw in Weak Stationarity that was only dictating conditions on 1st and 2nd moments.

Remark 1.10. Finite variance is not assumed for strong stationarity, but it is required for weak stationarity. Hence, strong stationarity does not imply weak stationarity.

- Strong $\not\Rightarrow$ Weak: Consider $Y_t \stackrel{\text{i.i.d.}}{\sim} \text{Cauchy}(0, 1)$. Because it is i.i.d., it is strictly stationary, but $\text{Var}(Y_t) = \infty$, so it is not weakly stationary.
- Weak $\not\Rightarrow$ Strong: Constant mean and variance do not guarantee that higher-order moments are independent of t .

However, for distributions completely characterized by their mean and variance (e.g., Gaussian processes), stationarity implies strong stationarity.

Proposition 1.11. If $\{Y_t\}$ is a Gaussian time series, then weak stationarity implies strong stationarity.

Example 1.12. Consider $\{Y_t\}_t$ where $Y_t = \epsilon_t$ and $\epsilon_t \sim \text{i.i.d.}(0, \sigma^2)$. This is stationary.

1. $\mathbb{E}[Y_t] = 0$
2. $\text{Var}(Y_t) = \sigma^2$
3. $\gamma_Y(t, s) = \mathbb{E}[Y_t Y_s] - \mu_t \mu_s = \mathbb{E}[\epsilon_t] \mathbb{E}[\epsilon_s] - 0 = 0$ (by independence).

Example 1.13. Consider $\{Y_t\}$ where $Y_t = \epsilon_t + \frac{1}{2}\epsilon_{t-1}$ and $\epsilon_t \sim \text{i.i.d.}(0, \sigma^2)$.

1. $\mathbb{E}[Y_t] = \mathbb{E}[\epsilon_t] + \frac{1}{2}\mathbb{E}[\epsilon_{t-1}] = 0$
2. $\text{Var}(Y_t) = \mathbb{E}[\epsilon_t^2 + \frac{1}{4}\epsilon_{t-1}^2 + \epsilon_t \epsilon_{t-1}] = \sigma^2 + \frac{\sigma^2}{4} + 0 = \frac{5}{4}\sigma^2$

3. Autocovariance for $t \neq s$:

$$\begin{aligned}
\gamma_Y(t, s) &= \mathbb{E}[Y_t Y_s] - \mu_t \mu_s \\
&= \mathbb{E} \left[\epsilon_t \epsilon_s + \frac{1}{2} \epsilon_t \epsilon_{s-1} + \frac{1}{2} \epsilon_s \epsilon_{t-1} + \frac{1}{4} \epsilon_{t-1} \epsilon_{s-1} \right] \\
&= \begin{cases} \mathbb{E}[\epsilon_t^2] + 0 + 0 + \frac{1}{4} \mathbb{E}[\epsilon_{t-1}^2], & t = s \\ 0 + \frac{1}{2} \mathbb{E}[\epsilon_{t-1}^2] + 0 + 0, & |t - s| = 1 \text{ (WLOG, } s = t + 1) \\ 0, & |t - s| \geq 2 \end{cases} \\
&= \begin{cases} \sigma^2 \left(1 + \frac{1}{4}\right), & t = s \\ \frac{\sigma^2}{2}, & |t - s| = 1 \\ 0, & |t - s| \geq 2 \end{cases}
\end{aligned}$$

This is stationary. Notice that $\text{Cov}(Y_t, Y_s) = 0$ only when s and t are at least two steps apart. This is natural since Y_t was defined in terms of the previous step. So, a one-step gap having some influence is expected.

Example 1.14. Consider the sum of error process $Y_t = \sum_{j=1}^t \epsilon_j$ for $\epsilon_t \sim \text{i.i.d.}(0, \sigma^2)$.

- $\mathbb{E}[Y_t] = 0$
- $\text{Var}(Y_t) = \mathbb{E}[Y_t^2] = \sum_{j \neq l} \mathbb{E}[\epsilon_j] \mathbb{E}[\epsilon_l] + \sum_{j=1}^t \mathbb{E}[\epsilon_j^2] = t\sigma^2$ (not free from t).

Naturally, a process where error accumulates with time depends on how much time has passed.

Example 1.15. $Y_t = a + bt + \epsilon_t$ where a, b are constants and $\epsilon_t \sim \text{i.i.d.}(0, \sigma^2)$.

- $\mathbb{E}[Y_t] = a + bt + \mathbb{E}[\epsilon_t] = a + bt$ (not free from t).

Example 1.16. $Y_t = A \cos(\theta t) + B \sin(\theta t)$ where A and B are uncorrelated random variables with zero mean and unit variance, and $\theta \in [-\pi, \pi]$.

- $\mathbb{E}[Y_t] = 0$
- $\mathbb{E}[Y_t^2] = \mathbb{E}[A^2 \cos^2(\theta t) + B^2 \sin^2(\theta t)] \leq A^2 + B^2 \leq 2 < \infty$
- Autocovariance:

$$\begin{aligned}
\gamma_Y(t, s) &= \mathbb{E}[A^2 \cos(\theta t) \cos(\theta s) + AB(\cos(\theta t) \sin(\theta s) + \sin(\theta t) \cos(\theta s)) + B^2 \sin(\theta t) \sin(\theta s)] \\
&= \cos(\theta t) \cos(\theta s) \mathbb{E}[A^2] + (\cos(\theta t) \sin(\theta s) + \sin(\theta t) \cos(\theta s)) \mathbb{E}[AB] + \sin(\theta t) \sin(\theta s) \mathbb{E}[B^2] \\
&= \cos(\theta t) \cos(\theta s) + \sin(\theta t) \sin(\theta s) \quad (\text{since } \text{Var}(A) = \text{Var}(B) = 1 \text{ and } \text{Cov}(A, B) = 0) \\
&= \cos(\theta(t - s))
\end{aligned}$$

Since $\gamma_Y(t, s)$ is a function of $(t - s)$ only, this process is weakly stationary.

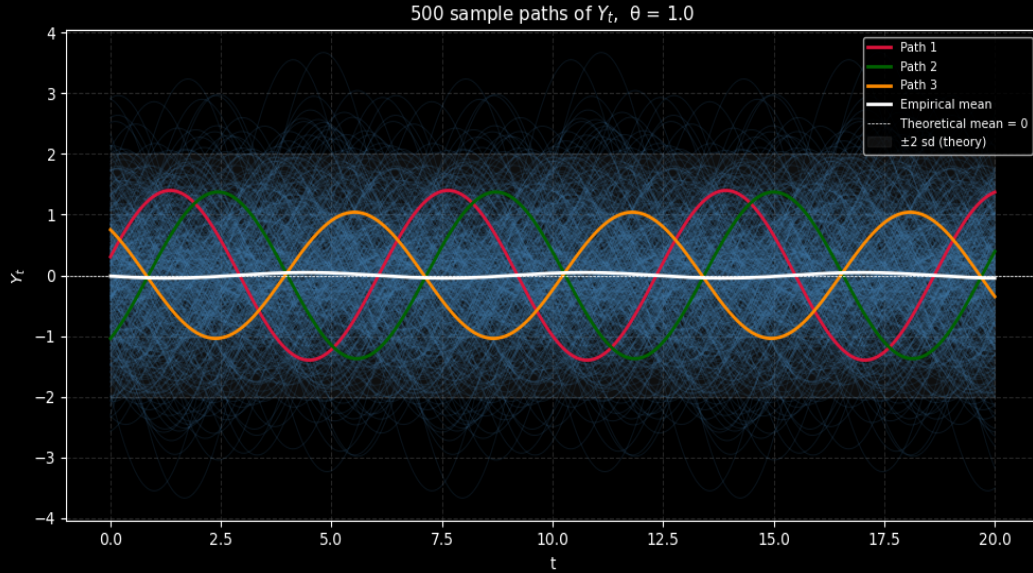


Figure 1.3: $Y_t = A \cos(\theta t) + B \sin(\theta t)$ where $A, B \sim \mathcal{N}(0, 1)$

Since we picked r.v. A and B to have standard normal distribution, as you increase number of simulated paths we can see them accumulating at 0, which is because A and B are very likely to be 0.

Example 1.17. Let $\{Z_t\}$ be centered i.i.d. random variables with constant variance σ^2 . Define

$$Y_t = Z_t + \theta Z_{t-1}$$

- $\mathbb{E}[Y_t] = 0$
- $\mathbb{E}[Y_t^2] = \mathbb{E}[Z_t^2] + \theta^2 \mathbb{E}[Z_{t-1}^2] + 2\theta \mathbb{E}[Z_t Z_{t-1}] = \sigma^2 + \theta^2 \sigma^2 + 0 < \infty$
- Autocovariance:

$$\begin{aligned} \gamma_Y(t, s) &= \mathbb{E}[Z_t Z_s + \theta Z_t Z_{s-1} + \theta Z_s Z_{t-1} + \theta^2 Z_{t-1} Z_{s-1}] \\ &= \begin{cases} \sigma^2(1 + \theta^2), & t = s \\ \theta \sigma^2, & |t - s| = 1 \\ 0, & |t - s| \geq 2 \end{cases} \end{aligned}$$

It is weakly stationary. Here too we see that γ_Y is 0 only when we are atleast two steps apart because process is defined to be dependent on previous step.

This forms a class of processes called the AR(1) processes. Later on we will see AR(p) processes. Note, this is different from example 1.13. They are moving average (MA) processes. We will study them in details.

Proposition 1.18. For any measurable function g and $\{Y_t\}_t$ being a strongly stationary process, $\{g(Y_t)\}_t$ is also strongly stationary.

Proof. Let $E = (Y_{s_1}, \dots, Y_{s_j})$ and $E' = (Y_{s_1+k}, \dots, Y_{s_j+k})$. Also let $B = B_1 \times \dots \times B_j$ where B_i are open sets.

$$\mathbb{P}(g(E) \in B) = \mathbb{P}(E \in g^{-1}(B)) = \mathbb{P}(E' \in g^{-1}(B)) = \mathbb{P}(g(E') \in B)$$

since $g^{-1}(B)$ is measurable and $E \stackrel{d}{=} E'$. □

This is, however, not true for weakly stationary processes.

Example 1.19. We will construct a process $\{Y_t\}$ that has an i.i.d. distribution with finite 1st and 2nd moments but infinite 4th moment. Then consider the non-linear function $g(y) = y^2$ to get the non-stationary process $\{g(Y_t)\}$.

Define a random variable X by:

$$\mathbb{P}(X = n) = \frac{c}{n^4}, \quad \text{where } c = \left(\sum_{n=1}^{\infty} \frac{1}{n^4} \right)^{-1} \quad (\text{normalizing constant}).$$

Let $\{Y_t\}$ be such that $Y_t \stackrel{\text{i.i.d.}}{\sim} X$ for all t . Then:

- $\mathbb{E}[Y_t] = \mathbb{E}[X] = \sum n \frac{c}{n^4} = c \sum \frac{1}{n^3} < \infty$
- $\mathbb{E}[Y_t^2] = \mathbb{E}[X^2] = \sum n^2 \frac{c}{n^4} = c \sum \frac{1}{n^2} < \infty \implies \{Y_t\}$ is stationary.

However:

$$\mathbb{E}[Y_t^4] = \mathbb{E}[X^4] = \sum n^4 \frac{c}{n^4} = c \sum 1 = \infty$$

Then, $\{Y_t^2\}$ is not stationary because the variance of the Y_t^2 process is infinite.

Characterization of Autocovariance Functions

We will now focus on the properties of the autocovariance function (ACvF). In fact, we will see a characterization of ACvFs, and for that we will need a result from stochastic analysis concerning when a collection of finite-dimensional distributions (f.d.d.'s) corresponds to a particular stochastic process $\{X_t\}_{t \in T}$ with $T = \mathbb{Z}, \mathbb{N}$, or $\mathbb{R}$.

We know that knowing the probability law on a π -system means we can uniquely extend it to the σ -algebra generated by that π -system. But in order to define a probability law on a π -system given the f.d.d.'s only, we need to verify a “marginal condition”.

Theorem 1.20. Kolmogorov Extension Theorem : Consider the family of probability distributions:

$$\{F_{t_1, \dots, t_n}(\cdot) : t_1, \dots, t_n \in \mathbb{Z}\}$$

These are the f.d.d.'s of a process if and only if for any $n \geq 1$, $t_1, \dots, t_n \in \mathbb{Z}$, and $1 \leq i \leq n$, we have:

$$\lim_{x_i \rightarrow \infty} F_{t_1, \dots, t_n}(x_1, \dots, x_n) = F_{\mathbf{t}(-i)}(\mathbf{x}(-i))$$

where $\mathbf{t}(-i)$ and $\mathbf{x}(-i)$ are $(n-1)$ -dimensional vectors obtained by removing the i -th component. Alternatively, since distributions and characteristic functions uniquely determine one another:

$$\lim_{u_i \rightarrow 0} \mathcal{C}_{t_1, \dots, t_n}(u_1, \dots, u_n) = \phi_{\mathbf{t}(-i)}(\mathbf{u}(-i))$$

Proof. Here we stick to constructing $X_t : \Omega \rightarrow E$ where $E = \mathbb{R}$ only.

Step 1. Pick sample space $\Omega = E^T$, and an element $\omega \in \Omega$ has the form $(\omega_t)_{t \in T}$ with $\omega_t \in E$.

Step 2. Define the process: $X_t(\omega) = \omega_t = \Pi_t(\omega)$ (projection of ω on the t -th coordinate).

Step 3. As always, since we cannot work with all subsets of E^T , we work with cylinder sets:

$$\{\omega : \omega_{t_1} \in A_1, \dots, \omega_{t_n} \in A_n\} = \bigcap_{j=1}^n \Pi_{t_j}^{-1}(A_j) =: S(t_1, \dots, t_n; A_1 \times \dots \times A_n)$$

for $n \in \mathbb{N}$ and $A_j \in \mathcal{B}(\mathbb{R})$.

Now, using the f.d.d.'s, we can assign probabilities to these sets. Let $F_{t_1, \dots, t_n}$ represent the measure $\mu_{t_1, \dots, t_n}$. Then define:

$$\mathbb{P}_0(S(t_1, \dots, t_n; A_1 \times \dots \times A_n)) = \mu_{t_1, \dots, t_n}(A_1 \times \dots \times A_n)$$

The role of the marginal condition is to ensure that the above definition of measure is well-defined. We may have two representations of the same cylinder set, and $\mathbb{P}_0$ should measure them equally. The only way a cylinder set has two representations is if we have a redundant coordinate that can take any value in E . For example:

$$S(1; A) = S(1, 2; A \times E)$$

The marginal condition simply says:

$$\mathbb{P}_0(S(1, 2; A \times E)) = \mu_{1,2}(A \times E) = \mu_1(A) = \mathbb{P}_0(S(1; A))$$

Now, we uniquely extend the law since the algebra formed by the cylinder sets generates the full product σ -algebra. $\square$

Proposition 1.21. Properties of ACvF : Since autocovariances are inner products on the space of centered random variables, the following properties follow:

1. $\gamma(0) := \gamma_Y(t, t) = \text{Cov}(Y_t, Y_t) = \text{Var}(Y_t) \geq 0$
2. $|\gamma(t, s)| = |\text{Cov}(Y_t, Y_s)| \leq \sqrt{\text{Var}(Y_t)} \sqrt{\text{Var}(Y_s)}$ (Cauchy-Schwarz inequality).
3. $\gamma(t, s) = \gamma(s, t)$ for all $s, t \in T$. Furthermore, if Y is stationary:

$$\gamma(s - t, 0) = \gamma(s, t) = \gamma(t, s) = \gamma(t - s, 0) \implies \gamma(t, s) = \gamma(|t - s|, 0) \quad \forall t, s \in T.$$

4. Non-negative definiteness.

Definition 1.22. Non-negative Definiteness : A real-valued function $f : \mathbb{Z} \rightarrow \mathbb{R}$ is said to be non-negative definite if and only if for any $n \in \mathbb{N}$, and for all choices of $a_1, \dots, a_n \in \mathbb{R}$ and $t_1, \dots, t_n \in \mathbb{Z}$:

$$\sum_{i=1}^n \sum_{j=1}^n a_i a_j f(t_i - t_j) \geq 0$$

Proposition 1.23. Characterization of Autocovariance Function : A real-valued even function on $\mathbb{Z}$ is non-negative definite if and only if it is the ACvF of some stationary process.

Proof. ($\Leftarrow$) Take $a_1, \dots, a_n \in \mathbb{R}$ and $t_1, \dots, t_n \in \mathbb{Z}$. Note that:

$$\begin{aligned} 0 &\leq \text{Var} \left(\sum_{i=1}^n a_i Y_{t_i} \right) = \mathbb{E} \left[\left\{ \sum_{i=1}^n a_i (Y_{t_i} - \mathbb{E}[Y_{t_i}]) \right\}^2 \right] \\ &= \mathbb{E} \left[\sum_{i=1}^n \sum_{j=1}^n a_i a_j (Y_{t_i} - \mathbb{E}[Y_{t_i}]) (Y_{t_j} - \mathbb{E}[Y_{t_j}]) \right] \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(Y_{t_i}, Y_{t_j}) \end{aligned}$$

So, $\text{Cov}(Y_{t_i}, Y_{t_j}) = \gamma_Y(t_i, t_j)$ is $\mathbb{R}$ -valued even function on $\mathbb{Z}$ that is non-negative definite.

($\Rightarrow$) Denote the real-valued even function on $\mathbb{Z}$ by f . We need to show there exists a time series with $\gamma_Y(t, s) = f(t - s)$.

Fix $n \geq 1$ and $t_1 < t_2 < \dots < t_n$ in $\mathbb{Z}$. Let the joint distribution $F_{t_1, \dots, t_n}$ have characteristic function $\phi_{t_1, \dots, t_n}$ defined by:

$$\mathcal{C}_{t_1, \dots, t_n}(u_1, \dots, u_n) = \exp \left(-\frac{1}{2} \mathbf{u}^T K \mathbf{u} \right)$$

where $\mathbf{u} = (u_1, \dots, u_n)^T$ and $K = (f(t_i - t_j))_{i,j=1}^n$.

Observe that by this construction, $(Y_{t_1}, \dots, Y_{t_n}) \sim \mathcal{N}(\mathbf{0}, K)$. Recall that $\mathcal{N}_n(\mathbf{0}, K)$ is a valid n -dimensional Gaussian distribution provided the matrix K is positive semi-definite and symmetric ($K = K^T$). Since f is non-negative definite, K is a non-negative definite matrix; and since f is even, $K = K^T$. Hence, K is a valid covariance matrix, and the f.d.d.'s are known.

Checking Kolmogorov's marginal condition:

$$\mathcal{C}_{t_1, \dots, t_n}(u_1, \dots, u_n) = \exp \left(-\frac{1}{2} \sum_{i,j \neq l} u_i u_j f(t_i - t_j) - \frac{1}{2} \sum_{j \neq l} u_l u_j f(t_l - t_j) - \frac{1}{2} \sum_{i \neq l} u_i u_l f(t_i - t_l) - \frac{1}{2} u_l^2 f(0) \right)$$

Taking the limit as $u_l \rightarrow 0$:

$$\begin{aligned} \lim_{u_l \rightarrow 0} \mathcal{C}_{t_1, \dots, t_n}(u_1, \dots, u_n) &= \exp \left(-\frac{1}{2} \sum_{i,j \neq l} u_i u_j f(t_i - t_j) \right) \\ &= \exp \left(-\frac{1}{2} \mathbf{u}(-l)^T K(-l) \mathbf{u}(-l) \right) \\ &= \mathcal{C}_{\mathbf{t}(-l)}(\mathbf{u}(-l)) \end{aligned}$$

The l -th row and column vanish from K , and the l -th coordinate vanishes from $\mathbf{u}$ when $u_l \rightarrow 0$.

Therefore, the required process exists on index set. Furthermore:

1. $\mathbb{E}[Y_t] = 0$
2. $\mathbb{E}[Y_t^2] = K_{tt} = f(0) < \infty$
3. $\text{Cov}(Y_t, Y_s) = f(|t - s|)$

Hence, $\{Y_t\}_t$ is stationary. □

Non-Stationarity Patterns.

Almost all practical examples lack stationarity and exhibit distinct patterns. Once patterns appear, the process is no longer “free from t ”. For example, if a process has an upward linear trend, there is a t term present. The following three are the primary reasons for non-stationarity:

- **Trend**
- **Seasonality / Cyclicity** (repeating patterns)
- **Heteroscedasticity** (changing variance)

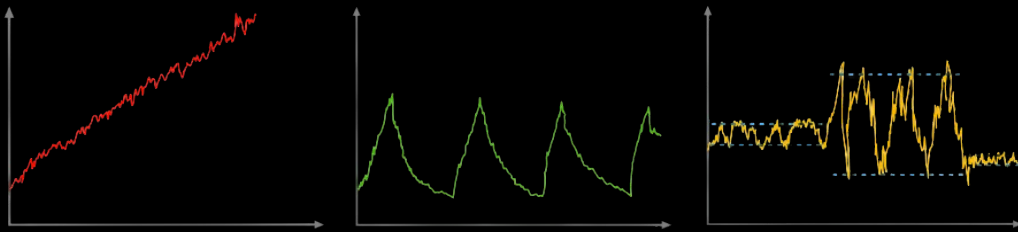


Figure 1.4: **Trend**, **Seasonality**, **Heteroscedasticity**

1.1.3 Time Series Decomposition

A time series can be separated into systematic components (predictable, caused by permanent causes) and noise:

- Secular trend (M_t) : Long-term movement or underlying direction of the time series.
- Seasonal component (S_t) : Repeating patterns occurring in fixed time intervals ($S_t = S_{t+s}$ for all t).
- Cyclical component (C_t) : Repeating patterns that fluctuate over irregular or unknown intervals.
- Noise: Random or irregular component (I_t)

Examples of Seasonal Fluctuation:

- Ice cream sales peak in summer and fall during winter.
- Money flow into daily commodity items increases at the beginning of each month after salaries are disbursed.
- Traffic volume is consistently high during morning office hours.

Examples of Cyclical Fluctuation:

- Sales decline during recessions and rise during economic recoveries, but the duration of a recession cycle is unknown.
- Number of Canadian lynx trapped each year: they enter regions during a particular season, but the cycle length and population peaks fluctuate irregularly over years.

Example 1.24 (Nottet Dataset). 20 years of monthly temperatures at Nottingham Castle in the UK. We clearly see periodic high and low patterns showing summers and winters over fixed intervals. This is purely seasonal with no trend and no cyclical component.

Once we remove the components M_t, S_t, C_t , what remains is I_t (the irregular component or noise). We are going to spend a lot of time in understanding the irregular component.

Decomposition Models

1. Additive Decomposition: Used when seasonal variations represent absolute amounts, e.g., a manufacturer produces 10,000 more units in November than in October.

$$Y_t = M_t + S_t + C_t + I_t \quad (1.4)$$

2. Multiplicative Decomposition: Used when seasonal variations represent relative percentage amounts, e.g., a manufacturer produces 20% more units in November than in October.

$$Y_t = M_t \cdot S_t \cdot C_t \cdot I_t \quad (1.5)$$

1.2 Basic models

Definition 1.25. White noise process $WN(0, \sigma^2)$: A process $\{\epsilon_t\}_{t \in T}$ is called a white noise process if

- (i) $\mathbb{E}[\epsilon_t] = 0$ for all $t \in T$,
- (ii) $\gamma_\epsilon(t, t+h) = \begin{cases} \sigma^2, & h = 0, \\ 0, & h \neq 0. \end{cases}$

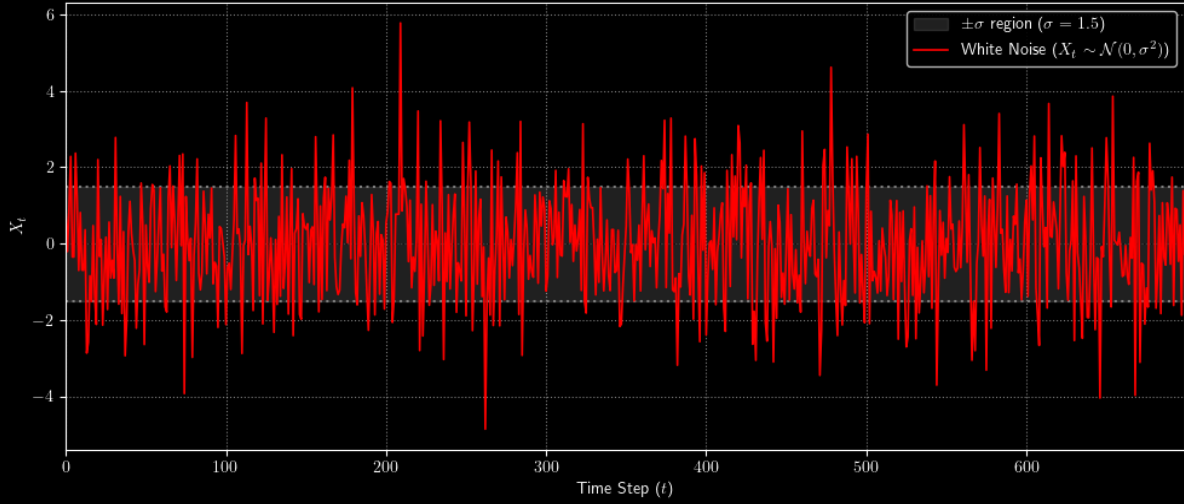


Figure 1.5: A single path of White noise process

Since we can see the figure, it is important to mention figure 1.6 is a linearly interpolated version of the actual WN process. Furthermore, sometimes $\{\epsilon_t\}_{t \in T}$ are $\text{iid}(0, \sigma^2)$, and independence gives us zero autocorrelation anyway. Due to these simplicities, $\text{WN}(0, \sigma^2)$ forms the building block for several more complicated stationary processes.

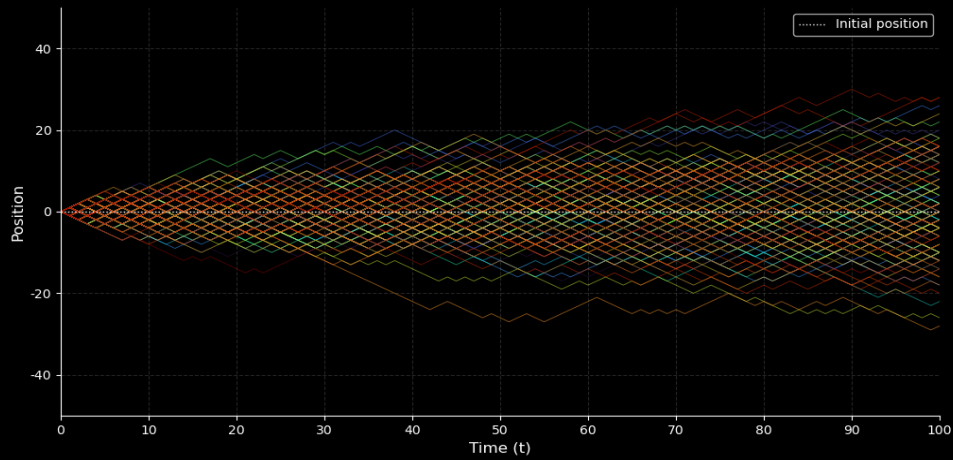


Figure 1.6: Simulation of 250 paths of a Random Walk process

Definition 1.26. Random walk process : Let $\{\epsilon_t\}$ be white noise. The random walk $\{Y_t\}$ is defined as

$$Y_t = \sum_{k \leq t} \epsilon_k = Y_{t-1} + \epsilon_t.$$

Definition 1.27. Linear trend process : Let $\{\epsilon_t\}_t$ be white noise. Then the linear trend process $\{Y_t\}_t$ is defined as

$$Y_t = a + bt + \epsilon_t.$$

We can think of this as there is an underlying linear process $a + bt$, and then at every step we have some error ϵ_t .

Definition 1.28. Moving average (MA) process : Let $\{\epsilon_t\}_t$ be white noise. Then the MA process $\{Y_t\}_t$ is defined as

$$Y_t = \epsilon_t + \frac{1}{2}\epsilon_{t-1}.$$

We are taking into account half the error of the previous step as well, on top of the current error. We see next the general version, where more errors are accounted for.

Definition 1.29. Moving average process of order $q \sim \text{MA}(q)$:

- Rather than using past values of the forecast, we use past errors. This makes the $\text{MA}(q)$ process very different from the $\text{AR}(p)$ processes that we are going to see next.
- The model uses q many previous errors. This helps in modelling the irregular component.

$$Y_t = c + \epsilon_t + \sum_{j=1}^q \theta_j \epsilon_{t-j}. \quad (1.6)$$

Often times we write,

$$Y_t = c + \sum_{j=0}^q \theta_j \epsilon_{t-j}, \quad \text{where } \theta_0 = 1,$$

and we get the above form again. This simply helps with the calculation.

Let us do the stationarity (etc.) checks for $\text{MA}(q)$ processes.

$$\begin{aligned} \mathbb{E}[Y_t] &= c, \\ \mathbb{E}[Y_t^2] &= c^2 + \sigma^2 + \sigma^2 \left(\sum \theta_j \right) < \infty. \end{aligned}$$

Moreover,

$$\begin{aligned} \gamma_Y(t, t+h) &= \mathbb{E}[(Y_t - c)(Y_{t+h} - c)] \\ &= \begin{cases} 0, & h > q, \\ \sigma^2 \sum_{i=0}^{q-h} \theta_i \theta_{i+h}, & 0 \leq h \leq q. \end{cases} \end{aligned}$$

A moving average process is hence indeed a weakly stationary process.

Calculations. With $\theta_0 = 1$,

$$\begin{aligned} (Y_t - c)(Y_{t+h} - c) &= \left(\sum_{j=0}^q \theta_j \epsilon_{t-j} \right) \left(\sum_{j=0}^q \theta_j \epsilon_{t+h-j} \right) \\ &= \sum_{i=0}^q \sum_{j=0}^q \theta_i \theta_j \epsilon_{t-i} \epsilon_{t+h-j}, \end{aligned}$$

and therefore

$$\gamma_Y(t, t+h) = \mathbb{E}[(Y_t - c)(Y_{t+h} - c)] = \sum_{i=0}^q \sum_{j=0}^q \theta_i \theta_j \underbrace{\mathbb{E}[\epsilon_{t-i} \epsilon_{t+h-j}]}_{\sigma^2 \text{ only if } t-i=t+h-j}.$$

Along with $j = i + h$ we have $j \leq q \Rightarrow i + h \leq q$. Let us see how the two sums collapse into one now that $j = i + h$. Fix some i and look at the inner sum running over j :

$$\sum_{j=0}^q \theta_j \mathbb{E}[\epsilon_{t-i} \epsilon_{t+h-j}] = \sum_{j=0}^q \theta_j \sigma^2 \mathbf{1}_{\{j=i+h\}} = \sigma^2 \sum_{j=0}^q \theta_j \mathbf{1}_{\{j=i+h\}} = \sigma^2 \begin{cases} \theta_{i+h}, & i+h \leq q, \\ 0, & i+h > q. \end{cases}$$

The index i was already running from 0 to q , and now it runs from 0 to $q - h$; the index j is gone. Hence

$$\Rightarrow \gamma_Y(t, t+h) = \sum_{i=0}^q \theta_i \left(\sigma^2 \theta_{i+h} \mathbf{1}_{\{i \leq q-h\}} \right) = \sigma^2 \sum_{i=0}^{q-h} \theta_i \theta_{i+h}.$$

Remark 1.30. $\gamma_Y(t, t+h)$ becomes 0 immediately after q and stays 0 for all $h > q$. So the plot of the ACvF of an MA(q) process tells us the value of q . In our diagram, for example, we see that q is 6. This will in fact give us the correct order of the MA process when we later on try to fit it to the data via the sample ACvF.

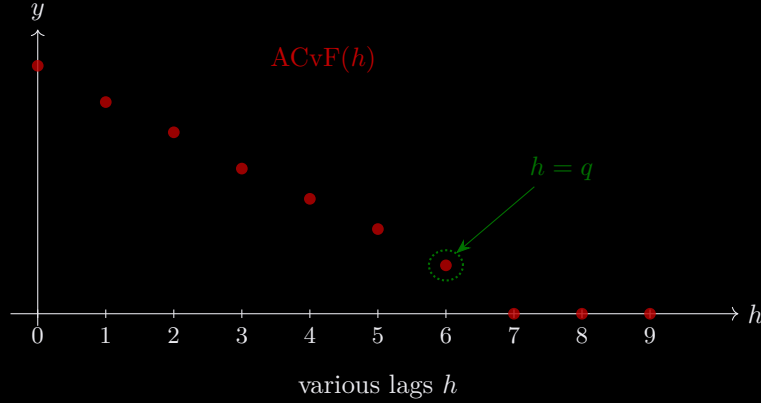


Figure 1.7: Autocovariance function of an MA(q) process: it vanishes for all lags $h > q$ (here $q = 6$).

Definition 1.31. Autoregressive process of order p , AR(p) :

- We forecast the variable of interest using a linear combination of past values. We call it autoregressive since it depends on its own past values.
- We call it of order p when it uses p -many past values.
- We involve an error (ϵ_t) and an intercept term (c). The intercept term is in the same essence as in polynomial regression, where it was the coefficient of the basis element 1.

$$Y_t = c + \epsilon_t + \sum_{j=1}^p \phi_j Y_{t-j}.$$

Basis interpretation.

$$Y_t = \underbrace{(c + \epsilon_t, \phi_1, \dots, \phi_p)}_{(p+1)\text{-dim vector}} \in \text{sp}\{1, Y_{t-1}, \dots, Y_{t-p}\},$$

which is the regression of Y_t on $\{Y_{t-1}, Y_{t-2}, \dots, Y_{t-p}\}$.

Mean and variance of AR(p) processes. MA(q) processes were always stationary. That is not true for AR(p) processes. However, we will see that AR and MA processes (and especially the ARMA processes that we are yet to see) are used to model stationary time series, and hence we will *assume* constant mean and variance for AR(p) processes. Keep in mind that we are assuming this and AR(p) process need not always be stationary.

Assume $\mathbb{E}[Y_{t-1}] = \dots = \mathbb{E}[Y_{t-p}] = \mu = \mathbb{E}[Y_t]$. Then,

$$\mu = \mathbb{E}[Y_t] = \mathbb{E}[c + \epsilon_t] + \sum_{j=1, \dots, p} \phi_j \mathbb{E}[Y_{t-j}] = c + \mu \left(\sum \phi_j \right) \Rightarrow \mu = \frac{c}{1 - \left(\sum_{j=1, \dots, p} \phi_j \right)}$$

Calculating $\text{Cov}(\cdot, \cdot)$ by brute-force calculation. Consider the demeaned process $X_t = Y_t - \mu$. Then

$$\begin{aligned} \Rightarrow X_t + \mu &= Y_t = c + \epsilon_t + \sum_{j=1}^p \phi_j Y_{t-j} = c + \epsilon_t + \sum_{j=1}^p \phi_j (X_{t-j} + \mu) \\ \Rightarrow X_t + \mu &= c + \epsilon_t + \sum_{j=1}^p \phi_j X_{t-j} + \mu \sum_{j=1}^p \phi_j \\ \Rightarrow X_t + \underbrace{\mu \left(1 - \left(\sum_{j=1}^p \phi_j \right) \right)}_{=c} &= c + \epsilon_t + \sum_{j=1}^p \phi_j X_{t-j}. \end{aligned}$$

Therefore the demeaned process $\{X_t\}_t$ satisfies the relation

$$X_t = \epsilon_t + \sum_{j=1}^p \phi_j X_{t-j}.$$

If $\{Y_t\}$ is stationary, then so is $\{X_t\} = \{Y_t - \mu\}$; also, $\text{Cov}(\cdot, \cdot)$ will be a function of the lag only for both. Then

$$\gamma_Y(h) = \text{Cov}(Y_{t-h}, Y_t) = \mathbb{E}[X_{t-h} X_t] = \gamma_X(h) = \sigma^2 \mathbf{1}_{\{h=0\}} + \sum_{j=1}^p \phi_j \gamma_X(|h-j|).$$

Calculation. We will do it in two cases, $h = 0$ and $h \neq 0$.

- Case $h \neq 0$. The trick is to not expand X_{t-h} at all. Expand X_t and multiply X_{t-h} to it. If we open both into their constituents, we will have many more terms to handle.

$$\mathbb{E}[X_{t-h} X_t] = \mathbb{E}\left[X_{t-h} \epsilon_t + \sum_{j=1}^p \phi_j X_{t-j} X_{t-h}\right] = \mathbb{E}[X_{t-h} \epsilon_t] + \sum_{j=1}^p \phi_j \mathbb{E}[X_{t-j} X_{t-h}].$$

Further, $\mathbb{E}[X_{t-h} \epsilon_t] = 0$ since X_{t-h} is uncorrelated with the future error ϵ_t . So a refinement of the trick is that we expand the X_s that comes later. For $\mathbb{E}[X_{t+h} X_t]$ we would expand X_{t+h} . Then

$$\begin{aligned} \text{Cov}(Y_{t-h}, Y_t) &= \sum_{j=1}^p \phi_j \gamma_X(t-j, t-h) = \sum_{j=1}^p \phi_j \gamma_X(|h-j|) \\ &\quad (\text{since } \{Y_t\} \text{ is stationary} \Rightarrow \{X_t\} \text{ is too}). \end{aligned}$$

- Case $h = 0$.

$$\mathbb{E}[X_t X_t] = \mathbb{E}[X_t \epsilon_t] + \sum_{j=1}^p \phi_j \mathbb{E}[X_{t-j} X_t] = \sigma^2 + \sum_{j=1}^p \phi_j \gamma_X(|h-j|).$$

Since $X_t \epsilon_t = \epsilon_t^2 + \sum \phi_j X_{t-j} \epsilon_t$, we get

$$\Rightarrow \underbrace{\mathbb{E}[X_t \epsilon_t]}_{=\sigma^2} = \sigma^2 + \sum \phi_j \cdot 0 \quad (\text{since } X_{t-j} \text{ is uncorrelated with } \epsilon_t).$$

Note that

$$\gamma_Y(h) = \sigma^2 \mathbf{1}_{\{h=0\}} + \sum_{j=1}^p \phi_j \gamma_Y(|h-j|).$$

The covariance of Y is in terms of the covariance of Y ! We pretty much came back to where we started.

Remark 1.32. This implicit representation, however, gives us a way to think too. Since $Y_t \in \text{sp}\{1, Y_{t-1}, \dots, Y_{t-p}\}$ for all t , we get

$$\begin{aligned} Y_{t-1} &\in \text{sp}\{1, Y_{t-2}, \dots, Y_{t-p-1}\}, \\ Y_{t-2} &\in \text{sp}\{1, Y_{t-3}, \dots, Y_{t-p-2}\}, \end{aligned}$$

and so on. We do iterative substitution. We look at $AR(1)$ in particular.

The $AR(1)$ process. Since we look back only one step, this is like a recursion. Starting from Y_0 ,

$$\begin{aligned} Y_1 &= c + \epsilon_1 + \phi_1 Y_0, \\ Y_2 &= c + \epsilon_2 + \phi_1 Y_1 = c + \epsilon_2 + \phi_1 (c + \epsilon_1 + \phi_1 Y_0) \\ &= c(1 + \phi_1) + \epsilon_2 + \phi_1 \epsilon_1 + \phi_1^2 Y_0, \\ Y_3 &= c + \epsilon_3 + \phi_1 Y_2 = c + \epsilon_3 + \phi_1 (c(1 + \phi_1) + \epsilon_2 + \phi_1 \epsilon_1 + \phi_1^2 Y_0) \\ &= c(1 + \phi_1 + \phi_1^2) + \epsilon_3 + \phi_1 \epsilon_2 + \phi_1^2 \epsilon_1 + \phi_1^3 Y_0, \\ &\vdots \\ Y_t &= c + \epsilon_t + \phi_1 Y_{t-1} = c + \epsilon_t + \phi_1 (c + \epsilon_{t-1} + \phi_1 Y_{t-2}) \\ &= c(1 + \phi_1) + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_1^2 Y_{t-2} \\ &\vdots \\ &= c(1 + \phi_1 + \dots + \phi_1^{t-1}) + \epsilon_t + \phi_1 \epsilon_{t-1} + \dots + \phi_1^{t-1} \epsilon_1 + \phi_1^t Y_0. \end{aligned}$$

If, instead of going all the way to Y_0 , we choose to stop at some k^{th} step, then we go back to $t - (k + 1)$. So the two representations are the following:

$$Y_t = \begin{cases} c(1 + \phi_1 + \phi_1^2 + \phi_1^3 + \dots + \phi_1^{t-1}) + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_1^2 \epsilon_{t-2} + \dots + \phi_1^{t-1} \epsilon_{t-(t-1)} + \phi_1^t Y_0, \\ c(1 + \phi_1 + \phi_1^2 + \dots + \phi_1^k) + \epsilon_t + \phi_1 \epsilon_{t-1} + \dots + \phi_1^k \epsilon_{t-k} + \phi_1^{k+1} Y_{t-(k+1)}. \end{cases}$$

Putting $k = t - 1$ in the second form gives us the first form again. Now, to study asymptotics, we notice that $|\phi_1| < 1$ is needed, since otherwise ϕ_1^{k+1} will blow up as $k \uparrow \infty$.

Remark 1.33. One would very naturally think that k is less than or equal to t , since we can only go back t many steps. This is not true in general, since we may assume $\{Y_t\}$ did not start at a finite time. In the usual real-life set-up that is the case, and we have $\{\dots, Y_{-1}, Y_0, Y_1, \dots\}$. So for us, $k \uparrow \infty \not\equiv t \uparrow \infty$; t is arbitrary but fixed.

Assume $|\phi_1| < 1$ (we will handle $|\phi_1| \geq 1$ later).

Then $|\phi_1^{k+1}| \rightarrow 0$ as $k \uparrow \infty$.

If $\{Y_t\}_t$ is to be stationary, we know $\text{Var}(Y_t) = \gamma_Y(t, t) = \gamma_Y(t - t, 0) = \gamma_Y(0)$ for all t . So,

$$\text{Var}(Y_{t-(k+1)}) = \mathbb{E}[(Y_{t-(k+1)})^2] - (\mathbb{E}[Y_{t-(k+1)}])^2$$

is constant too, hence free from k . Therefore

$$\begin{aligned} \Rightarrow \quad \mathbb{E}[Y_{t-(k+1)}^2] &= \text{Var}(Y_{t-(k+1)}) + (\mathbb{E}[Y_{t-(k+1)}])^2 \\ &= \text{Var}(Y_t) + (\mathbb{E}[Y_t])^2 \quad (\text{since } \{Y_t\} \text{ is stationary}) \\ &\equiv \text{constant}. \end{aligned}$$

Now,

$$\mathbb{E}\left[\left\{Y_t - c \sum_{j=0}^k \phi_1^j - \sum_{j=0}^k \phi_1^j \epsilon_{t-j}\right\}^2\right] = \mathbb{E}\left[\left\{\phi_1^{k+1} Y_{t-(k+1)}\right\}^2\right] = \underbrace{|\phi_1|^{2k+2}}_{\rightarrow 0} \underbrace{\mathbb{E}[Y_{t-(k+1)}^2]}_{\text{const.}}$$

Therefore

$$\left\| Y_t - \sum_{j=0}^k \phi_1^j (c + \epsilon_{t-j}) \right\|_{L^2} \rightarrow 0 \quad \text{if } |\phi_1| < 1 \text{ and } \{Y_t\}_t \text{ is stationary.} \quad (1.7)$$

So, $\left\{ \sum_{j=0}^k \phi_1^j (c + \epsilon_{t-j}) \right\}_k$ converges in the mean square sense to $\sum_{j=0}^{\infty} \phi_1^j (c + \epsilon_{t-j}) = Y_t$ for all t .

Our last remark plays a vital role in concluding the last line; since $k \uparrow \infty$ does not imply t blows up, Y_t is still a random variable at the finite time t . Now, more precisely,

$$Y_t = \frac{c}{1 - \phi_1} + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j}.$$

Under the assumption that $\{Y_t\}_t$ is stationary, the process defined recursively in terms of its previous step only has now become the net weighted sum of all previous errors added to the mean. This is also called the MA(∞) representation of AR(1).

We showed: $\{Y_t\}$ stationary and $|\phi_1| < 1 \Rightarrow$ MA(∞) representation. For the rest of the discussion we prove the other implication. The plan is:

1. Forget $Y_t :=$ MA(∞) form and simply show when an MA(∞) process is stationary.
2. Show that the MA(∞) process satisfies the AR(1) equation $Y_t = c + \epsilon_t + \phi_1 Y_{t-1}$.

Then 1. and 2. together give the existence of a stationary solution of the AR(1) equation. This is simply the converse of the above claim, which says that any stationary solution of AR(1) has an MA(∞) representation.

Consider a process

$$W_t = \sum_{j=0}^{\infty} \phi_1^j (c + \epsilon_{t-j}), \quad \text{where } |\phi_1| < 1. \quad (1.8)$$

- $\mathbb{E}[W_t] = c \sum_{j=0}^{\infty} \phi_1^j$ for all $t \in \mathbb{Z}$.
- $\text{Cov}(W_{t+h}, W_t) = \text{Cov}\left(\lim W_{t+h}^{(n)}, \lim W_t^{(n)}\right)$, where $W_s^{(n)} = \sum_{j=0}^n \phi_1^j (c + \epsilon_{s-j})$.

For any $s \in T$, $\{W_s^{(n)}\}_n$ is Cauchy in the $L^2(\mathbb{P})$ sense, and hence, by completeness, there exists a unique limit. We prove the Cauchy property of $\{W_s^{(n)}\}_n$ next.

$$\begin{aligned} \mathbb{E}\left[\{W_s^{(m)} - W_s^{(n)}\}^2\right] &= \mathbb{E}\left[\left\{\sum_{j=n+1}^m \phi_1^j (c + \epsilon_{t-j})\right\}^2\right] \\ &= \mathbb{E}\left[\sum_{i,j=n+1}^m \phi_1^i \phi_1^j (c + \epsilon_{t-i})(c + \epsilon_{t-j})\right] \\ &= \sum_{i,j=n+1}^m \phi_1^i \phi_1^j \mathbb{E}[(c + \epsilon_{t-i})(c + \epsilon_{t-j})] \\ &= \sum_{i,j=n+1}^m \phi_1^i \phi_1^j \mathbb{E}[c^2 + c\epsilon_{t-j} + c\epsilon_{t-i} + \epsilon_{t-i}\epsilon_{t-j}] \\ &= \sum_{i,j=n+1}^m \phi_1^i \phi_1^j c^2 + \sum_{i=n+1}^m \phi_1^{2i} \mathbb{E}[\epsilon_{t-i}^2] \quad (\text{cross terms vanish}) \\ &= c^2 \sum_{i+j=\ell=n+2}^{2m} \phi_1^\ell + \sigma^2 \sum_{i=n+1}^m (\phi_1)^{2i} \rightarrow c^2 \cdot 0 + \sigma^2 \cdot 0 = 0. \end{aligned}$$

Since $|\phi_1| < 1$, we have $\sum_0^\infty \phi_1^j < \infty$, and $\sum_{n+2}^{2m} \phi_1^\ell$ and $\sum_{n+1}^m \phi_1^{2i}$ are tails of a convergent infinite series. Hence they tend to 0. Therefore

$$\mathbb{E}\left[\{W_s^{(m)} - W_s^{(n)}\}^2\right] \rightarrow 0 \quad \text{as } n, m \rightarrow \infty,$$

and the limit of $W_s^{(m)}$ in the $L^2(\mathbb{P})$ sense naturally is W_s . Then,

$$\begin{aligned} W_s^{(n)} &\xrightarrow{L^2} W_s \quad \forall s \in T \Rightarrow \mathbb{E}[W_t^{(n)} W_s^{(n)}] \rightarrow \mathbb{E}[W_t W_s] \\ (\because \text{we only subtract a constant}) &\iff \text{Cov}(W_t^{(n)}, W_s^{(n)}) \rightarrow \text{Cov}(W_t, W_s) \end{aligned}$$

The following lemma is a result from Measure theory and used in the last claims.

Lemma 1.34. *If $X_n \xrightarrow{L^2} X$ and $Y_n \xrightarrow{L^2} Y$, then $\mathbb{E}[X_n Y_n] \rightarrow \mathbb{E}[XY]$.*

Proof. Write $X_n Y_n - XY = (X_n - X)Y_n + X(Y_n - Y)$. Then

$$\begin{aligned} \mathbb{E}[X_n Y_n - XY] &= \mathbb{E}[(X_n - X)Y_n] + \mathbb{E}[X(Y_n - Y)] \\ &\leq \underbrace{\|X_n - X\|_{L^2}}_{\rightarrow 0} \underbrace{\|Y_n\|_{L^2}}_{< \infty} + \underbrace{\|X\|_{L^2}}_{< \infty} \underbrace{\|Y_n - Y\|_{L^2}}_{\rightarrow 0} \rightarrow 0. \end{aligned}$$

Here $X \in L^2$ since $X_n \rightarrow X$ in L^2 and L^2 is complete. □

Alternate proof. This argument is surprisingly easy.

$$\lim_n \text{Cov}(W_t^{(n)}, W_s^{(n)}) = \text{Cov}\left(\lim W_t^{(n)}, \lim W_s^{(n)}\right) = \text{Cov}(W_t, W_s).$$

This is because inner products are continuous, so $\text{Cov}(X_n, Y_n) \rightarrow \text{Cov}(X, Y)$. □

All the calculation so far was to exchange the limit; now we simply calculate $\text{Cov}(W_t^{(n)}, W_s^{(n)})$.

$$\begin{aligned} \text{Cov}(W_{t+h}^{(n)}, W_t^{(n)}) &= \mathbb{E}\left[\sum_{i=0}^n \sum_{j=0}^n \phi_1^i \phi_1^j \epsilon_{t+h-i} \epsilon_{t-j}\right] \\ &= \sum_{j=0}^n \phi_1^{h+j} \phi_1^j \mathbb{E}[\epsilon_{t-j}^2] \quad (t+h-i = t-j \Rightarrow i = h+j) \\ &= \sigma^2 \phi_1^h \sum_{j=0}^n \phi_1^{2j}. \end{aligned}$$

Therefore

$$\text{Cov}(W_{t+h}, W_t) = \frac{\sigma^2 \phi_1^h}{1 - \phi_1^2} \quad (\text{geometric series}).$$

So the mean is constant and the ACvF is a function of the lag h only. The $\text{MA}(\infty)$ process $\sum_{j=0}^\infty \phi_1^j (c + \epsilon_{t-j})$ from 1.8 is stationary if and only if the weight satisfies $|\phi_1| < 1$.

We next see that W_t satisfies $Y_t = c + \epsilon_t + \phi_1 Y_{t-1}$.

$$\begin{aligned} W_{t-1} &= \sum_{j=0}^\infty \phi_1^j (c + \epsilon_{t-1-j}) = (c + \epsilon_{t-1}) + \phi_1 (c + \epsilon_{t-2}) + \dots \\ \Rightarrow \phi_1 W_{t-1} &= \sum_{j=0}^\infty \phi_1^{j+1} (c + \epsilon_{t-1-j}) = \phi_1 (c + \epsilon_{t-1}) + \phi_1^2 (c + \epsilon_{t-2}) + \dots \\ \Rightarrow c + \epsilon_t + \phi_1 W_{t-1} &= c + \epsilon_t + \phi_1 (c + \epsilon_{t-1}) + \phi_1^2 (c + \epsilon_{t-2}) + \dots := W_t \quad (\text{by definition}). \end{aligned}$$

Proposition 1.35. Consider the AR(1) equation $Y_t = c + \epsilon_t + \phi Y_{t-1}$. If $|\phi| < 1$, then there exists a unique solution of the above equation of the form

$$\text{MA}(\infty) := \sum_{j=0}^{\infty} \phi^j (c + \epsilon_{t-j}).$$

What if $|\phi_1| \geq 1$?

The very first limit argument fails when $|\phi_1| \geq 1$, since $|\phi_1|^{2k+1} \rightarrow 0$ was needed. In the case $|\phi_1| \geq 1$ we will first look only at the case $|\phi_1| > 1$. The natural way to exploit this is to involve ϕ_1^{-1} :

$$\phi_1^{-1} Y_t = \phi_1^{-1} (c + \epsilon_t) + Y_{t-1} \Rightarrow Y_t = \phi_1^{-1} Y_{t+1} - \phi_1^{-1} (c + \epsilon_{t+1}), \quad \text{where } |\phi_1^{-1}| < 1.$$

Let $\tilde{\epsilon}_t = -\phi_1^{-1} \epsilon_t$, where $\text{Var}(\tilde{\epsilon}_t) = \frac{\sigma^2}{\phi_1^2} := \tilde{\sigma}^2$. So $\{\tilde{\epsilon}_t\}_t$ is a new $\text{WN}(0, \tilde{\sigma}^2)$. Also let $-\frac{c}{\phi_1} = \tilde{c}$. Then the modified AR(1) process is

$$Y_t = (\tilde{c} + \tilde{\epsilon}_{t+1}) + \phi_1^{-1} Y_{t+1}.$$

The way to solve this is exactly like earlier: we do iterative substitution and then use measure-theoretic arguments. We will, however, simply exploit the available solutions. We have

$$Y_t = c \sum_{j=0}^k \phi_1^j + \sum_{j=0}^k \phi_1^j \epsilon_{t-j} + \phi_1^{k+1} Y_{t-(k+1)}.$$

Divide by ϕ_1^{k+1} :

$$\begin{aligned} \frac{Y_t}{\phi_1^{k+1}} &= c \left(\frac{1}{\phi_1^{k+1}} + \cdots + \frac{1}{\phi_1} \right) + \frac{\epsilon_t}{\phi_1^{k+1}} + \frac{\epsilon_{t-1}}{\phi_1^k} + \cdots + \frac{\epsilon_{t-k}}{\phi_1} + Y_{t-(k+1)} \\ \Rightarrow Y_{t-(k+1)} &= \frac{Y_t}{\phi_1^{k+1}} - c \left(\frac{1}{\phi_1^{k+1}} + \cdots + \frac{1}{\phi_1} \right) - \left(\frac{\epsilon_t}{\phi_1^{k+1}} + \frac{\epsilon_{t-1}}{\phi_1^k} + \cdots + \frac{\epsilon_{t-k}}{\phi_1} \right). \end{aligned}$$

The goal is to make Y_s the subject; hence we put $t \sim t + (k+1)$:

$$\begin{aligned} Y_t &= (\phi_1^{-1})^{k+1} Y_{t+(k+1)} - c \sum_{i=1}^{k+1} (\phi_1^{-1})^i - \left(\frac{\epsilon_{t+(k+1)}}{\phi_1^{k+1}} + \frac{\epsilon_{t+k}}{\phi_1^k} + \cdots + \frac{\epsilon_{t+1}}{\phi_1} \right) \\ \Rightarrow Y_t &= (\phi_1^{-1})^{k+1} Y_{t+(k+1)} - c \sum_{i=1}^{k+1} (\phi_1^{-1})^i - \sum_{i=1}^{k+1} (\phi_1^{-1})^i \epsilon_{t+i} \\ \Rightarrow Y_t &= (\phi_1^{-1})^{k+1} Y_{t+(k+1)} + \tilde{c} \sum_{i=0}^k (\phi_1^{-1})^i + \sum_{i=0}^k (\phi_1^{-1})^i \tilde{\epsilon}_{t+i}. \end{aligned}$$

Now, like before,

$$\left\| Y_t - \tilde{c} \sum_{i=0}^k \phi_1^{-i} - \sum_{i=0}^k (\phi_1^{-1})^i \tilde{\epsilon}_{t+i} \right\|_2^2 = \left\| (\phi_1)^{-(k+1)} Y_{t+(k+1)} \right\|_2^2 \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

This gives us the solutions we have up till now.

Proposition 1.36 (Stationary solutions of AR(1)).

$$Y_t = \begin{cases} \sum_{i=0}^{\infty} (c + \epsilon_{t-i}) \phi_1^i = \frac{c}{1 - \phi_1} + \sum_{i=0}^{\infty} \phi_1^i \epsilon_{t-i}, & |\phi_1| < 1, \\ -\sum_{i=1}^{\infty} (c + \epsilon_{t+i}) \phi_1^{-i} = \frac{c}{1 - \phi_1} - \sum_{i=1}^{\infty} \phi_1^{-i} \epsilon_{t+i}, & |\phi_1| > 1. \end{cases}$$

For $|\phi_1| < 1$, the present Y_t depends on the past noise $\{\epsilon_s : s < t, s \in \mathbb{Z}\}$.
For $|\phi_1| > 1$, the present Y_t depends on the future noise $\{\epsilon_s : s > t, s \in \mathbb{Z}\}$.

Remark 1.37. We will naturally focus on the case $|\phi_1| < 1$ for modelling, since the case $|\phi_1| > 1$ leads us to depend on the future, and that is practically not feasible.

Question 1.38. One might ask: by doing so, are we not missing out on models that are actually available in nature?

If such a process does exist in nature, then, mathematically put, it is essentially saying that the equation

$$Y_t = \phi_1 Y_{t-1} + c + \epsilon_t, \quad \{\epsilon_t\}_t \sim \text{WN}(0, \sigma^2), \quad |\phi_1| > 1,$$

has a stationary solution. Then, by the ways already used in the above proof, it can be shown that $\{Y_t\}$ satisfies the following AR(1) equation as well, for an appropriate τ :

$$Y_t = \phi_1^{-1} Y_{t-1} + \tilde{\epsilon}_t, \quad \epsilon_t \sim \text{WN}(0, \tau^2).$$

So nothing is mathematically lost by eliminating $|\phi_1| > 1$.

Lastly, what if $|\phi_1| = 1$? There does not exist any stationary solution of the AR(1) equation.

Let, if possible, a stationary process $\{Y_t\}$ satisfy the AR(1) equation. Let the mean and variance of this stationary process be the constants μ and s^2 . Then

$$\begin{aligned} s^2 &= \text{Var}(Y_t) = \text{Var}(\phi_1 Y_{t-1} + \epsilon_t) = \phi_1^2 \text{Var}(Y_{t-1}) + \text{Var}(\epsilon_t) = \phi_1^2 s^2 + \sigma^2 \\ \Rightarrow s^2(1 - \phi_1^2) &= \sigma^2. \end{aligned}$$

Since $|\phi_1| = 1 \Rightarrow \phi_1^2 = 1 \Rightarrow \sigma^2 = 0!!$

The AR(2) process.

$$\begin{aligned} Y_t &= c + \epsilon_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2}, \\ \{Y_t\} &= \{Y_0, Y_1, c + \epsilon_2 + \phi_1 Y_1 + \phi_2 Y_0, c + \epsilon_3 + \phi_1 Y_2 + \phi_2 Y_1, \dots\}. \end{aligned}$$

Remark 1.39. To find stationarity (which we will do later) for AR(p) models where $p > 2$, the calculations get considerably hard. We use hypothesis testing etc. to do them.

Notational fixes. Define the following two $\mathbb{C}$ -polynomials first:

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \dots - \phi_p z^p; \quad \Theta(z) = 1 + \theta_1 z + \theta_2 z^2 + \dots + \theta_q z^q, \quad z \in \mathbb{C}.$$

Define the backward difference operator B by

$$B^j Z_t = Z_{t-j} \quad (\text{a linear operator}).$$

Recall

$$\begin{aligned} \text{MA}(q) : \quad Y_t &= c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q}, \\ \text{AR}(p) : \quad Y_t &= c + \epsilon_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} \\ \Rightarrow \quad \epsilon_t &= -c + Y_t - \phi_1 Y_{t-1} - \dots - \phi_p Y_{t-p}. \end{aligned}$$

Hence

$$\begin{aligned} \text{MA}(q) &:= Y_t = c + \Theta(B)\epsilon_t, \\ \text{AR}(p) &:= \epsilon_t = -c + \phi(B)Y_t. \end{aligned}$$

Definition 1.40 (Autoregressive moving average process, ARMA(p, q)). • It simply combines weighted p -many previous values and weighted q -many previous errors:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \epsilon_t + \sum_{i=1}^q \theta_i \epsilon_{t-i}.$$

Alternatively,

$$Y_t - \sum_{i=1}^p \phi_i Y_{t-i} = c + \epsilon_t + \sum_{i=1}^q \theta_i \epsilon_{t-i} \sim \phi(B)Y_t = c + \Theta(B)\epsilon_t.$$

- We will assume ARMA(p, q) to be stationary as part of the definition. This is because ARMA processes are used to model stationary processes; to handle non-stationary ones we have other ways. Just keep in mind that we are assuming the stationarity, and not all ARMA(p, q) processes need be stationary. It is very easy to construct an example using AR(1) where $|\phi_1| = 1$:

$$Y_t = 1 + Y_{t-1} + \epsilon_t \quad (\theta_i = 0 \quad \forall i, \quad \phi_j = 0 \quad \forall j \notin \{1\}, \quad \phi_1 = 1).$$

So, simply adjusting the values $\{\theta_i\}_i$ and $\{\phi_j\}_j$ in ARMA(p, q) can create any MA(q) and AR(p) process.

Definition 1.41 (Linear filters). Suppose we have a time series $\{\dots, X_{-1}, X_0, X_1, \dots\}$, and suppose $\{Y_t\}_t$ is another time series built using a linear combination of past, present and future values of $\{X_t\}_t$. Then the *linear filter* is the mathematical operation (transformation) that creates $\{Y_t\}_t$ from $\{X_t\}_t$.

Definition 1.42 (Causal). An ARMA(p, q) process is called *causal*, or a causal function of $WN(0, \sigma^2)$, if there exists a sequence of constants $\{\psi_j\}_{j \in \mathbb{N}_0}$ with $\sum_j |\psi_j| < \infty$ and

$$Y_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}, \quad t \in \mathbb{Z}.$$

The name causal (causality) is due to the fact that the process right now depends on the entire past and present white noise process: $\{\psi_j\}_{j \in \mathbb{N}_0}$ and $\{\epsilon_s\}_{s=-\infty}^t$ “cause” Y_t to happen.

Naturally, $\{\psi_j\}_j$ being 0s would mean past noises having no effect. Nevertheless, the point that the future does not interfere still stands. We will be very much interested in “causal” models only.

Key classifications of linear filters.

1. *Causal and non-causal.* We call it causal when it depends on the past and the present. We call $\{Y_t\}$ non-causal when the dependence is on the future too.
2. *FIR and IIR filters.* FIR stands for finite impulse response, and this is when we look at a finite linear combination, like the MA(q) process, where only q -many terms are involved. IIR stands for infinite impulse response, like MA(∞) etc.

Example 1.43.

$$Y_t = \sum_{j=0}^{\infty} \psi_j X_{t-j} \equiv \Psi(B)X_t \sim \text{causal IIR filters.}$$

In our analysis we will often write such models as $\Psi(B)\epsilon_t$, where $\Psi(z) = \sum_{j=0}^{\infty} \psi_j z^j$, $z \in \mathbb{C}$. Let us see how things change when we move from $\Psi(z)$ to $\Psi(B)$ and vice versa. The expression $\sum_{j=0}^{\infty} \psi_j z^j$ is a power series, and $\sum_{j=0}^{\infty} \psi_j B^j$ is an operator.

For a power series we need to ensure what the domain of convergence is, since otherwise it has no meaning. For the latter, we only know that B and B^j are operators for any $j \in \mathbb{N}$. The infinite sum need not be a valid operator. The next result discusses when $\Psi(B)$ makes sense.

Proposition 1.44. *If $\{X_t\}_t$ is any sequence of random variables such that $\sup_t \mathbb{E}|X_t| < \infty$, and if $\sum_{i=-\infty}^{\infty} |\psi_i| < \infty$, then the following series converges absolutely almost surely:*

$$\Psi(B)X_t := \sum_{j=-\infty}^{\infty} \psi_j X_{t-j}.$$

In addition, if $\sup_t \mathbb{E}[X_t^2] < \infty$, then it also converges in mean square.

This is essentially saying when it makes sense to talk about these infinite causal sums. The requirement is that the supremum of the absolute means is finite and the coefficients are absolutely summable.

Remark 1.45. *This is true for any sequence of random variables $\{X_t\}_t$; no independence structure is required.*

Proof. Let $M = \sup_t \mathbb{E}|X_t|$ and $C = \sum_{-\infty}^{\infty} |\psi_i|$. Define $S_N = \sum_{-N}^N |\psi_j X_{t-j}|$. Then

$$\begin{aligned} \mathbb{E}[S - S_n] &= \left\| \sum_{|j|>n} |\psi_j X_{t-j}| \right\|_{L^1} \leq \sum_{|j|>n} \|\psi_j X_{t-j}\|_{L^1} \quad (\Delta\text{-inequality}) \\ &= \sum_{|j|>n} |\psi_j| \|X_{t-j}\|_{L^1} \leq M \sum_{|j|>n} |\psi_j| \longrightarrow M \cdot 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Moreover, let $\sup_t \mathbb{E}[X_t^2] = K$. (The same trick as above will work for any L^p , in fact.) Then

$$\mathbb{E}[|S - S_n|^2] = \left\| \sum_{|j|>n} |\psi_j X_{t-j}| \right\|_{L^2}^2 \leq \sum_{|j|>n} |\psi_j|^2 \|X_{t-j}\|_{L^2}^2 < \sqrt{K} \sum_{|j|>n} |\psi_j| \longrightarrow \sqrt{K} \cdot 0 \quad \text{as } n \rightarrow \infty.$$

□

Corollary 1.46. *If $\{X_t\}_t$ is a stationary process with finite mean, finite variance and $\gamma_X(\cdot)$ as the ACvF, and if $\sum_{-\infty}^{\infty} |\psi_i| < \infty$, then for each $t \in \mathbb{Z}$ the series $\Psi(B)X_t$ converges absolutely almost surely and in mean square.*

Also, if we call $\Psi(B)X_t := V_t$, then $\{V_t\}$ is stationary with ACvF given by

$$\gamma_V(h) = \sum_{j,k=-\infty}^{\infty} \psi_j \psi_k \gamma_X(h - j + k).$$

Proof. The corollary follows since stationary processes have constant mean and variance. First,

$$\mathbb{E}[V_t] = \mathbb{E}\left[\sum_{-\infty}^{\infty} \psi_j X_{t-j}\right] = \sum_{-\infty}^{\infty} \psi_j \mathbb{E}[X_{t-j}] \quad (\text{linearity}) = \mu_X \sum_{-\infty}^{\infty} \psi_j = \tilde{C}\mu_X,$$

where $|\sum \psi_j| \leq \sum |\psi_j| < \infty$. So the mean is constant, μ_V . Next,

$$\begin{aligned} \text{Cov}(V_{t+h}, V_t) &= \mathbb{E}[(V_{t+h} - \mu_V)(V_t - \mu_V)] = \mathbb{E}\left[\left(\sum_{-\infty}^{\infty} \psi_j (X_{t+h-j} - \mu_X)\right)\left(\sum_{-\infty}^{\infty} \psi_k (X_{t-k} - \mu_X)\right)\right] \\ &= \sum_{j,k=-\infty}^{\infty} \psi_j \psi_k \underbrace{\mathbb{E}[(X_{t+h-j} - \mu_X)(X_{t-k} - \mu_X)]}_{\text{covariance of } X \text{ with lag } (t+h-j)-(t-k)=h-j+k} \\ &= \sum_{j,k=-\infty}^{\infty} \psi_j \psi_k \gamma_X(h - j + k). \end{aligned}$$

□

While interchanging the power series $\Psi(z)$ and the operator $\Psi(B)$, recall that we ran into two problems: the domain of convergence and the domain of definition. In the next proof we will work a lot to show

$$\phi(B)\Psi(B) = \Theta(B) \Rightarrow \phi(z)\Psi(z) = \Theta(z),$$

and also how

$$\Psi(z) = \frac{\Theta(z)}{\phi(z)} \Rightarrow \Psi(B) = \frac{\Theta(B)}{\phi(B)}.$$

They do not “just follow” and genuinely require a proof.

Remark 1.47. Also recall, for a linear operator T and power series f and g (on their defined domains):

- $f(z) = g(z) \Rightarrow f(T) = g(T)$? This holds only when $f(T)$ and $g(T)$ are defined.
- $f(T) = g(T) \Rightarrow f(z) = g(z)$? This is not true in general.

(A projection has the property $P^2 = P$. Let $f(z) = z^2$ and $g(z) = z$. Clearly $f(P) = g(P)$ but $f \neq g$.)

Theorem 1.48 (Characterising causal ARMA(p, q)). Let $\{Y_t\}_t$ be an ARMA(p, q) process of the form $\phi(B)Y_t = \Theta(B)\epsilon_t$ for $\{\epsilon_t\} \sim \text{WN}(0, \sigma^2)$. Suppose the polynomials $\phi(\cdot)$ and $\Theta(\cdot)$ have no common zeros. Then

$$\{Y_t\} \text{ is a causal ARMA}(p, q) \iff \phi \text{ has no zeros in the unit disc } \mathbb{D} \subseteq \mathbb{C}.$$

The coefficients $\{\psi_j\}$ in the expansion $Y_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}$ can be determined by

$$\Psi(z) = \sum_{j=0}^{\infty} \psi_j z^j = \frac{\Theta(z)}{\phi(z)}, \quad z \in \mathbb{D}.$$

Moreover, $\{Y_t\}$ defined as above is the unique stationary (and causal) solution time series of the ARMA(p, q) process of the form $\phi(B)Y_t = \Theta(B)\epsilon_t$.

Proof. Let us take care of the very last part first. “Unique” holds since $Y_t = \sum \psi_j \epsilon_{t-j}$, where the coefficients ψ_j are uniquely determined by the power series $\Theta(z)/\phi(z)$, $z \in \mathbb{D}$, for the Θ, ϕ given in ARMA(p, q).

Stationary and causal will follow once we establish that Y_t has the form $\sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}$.

Lastly, we show that $Y_t = \sum \psi_j \epsilon_{t-j}$ is a solution to $\phi(B)Y_t = \Theta(B)\epsilon_t$. This will actually be very easy once we are done with the proof, since all we need is to exchange the power series for the operator; the entire proof is essentially a justification for doing that exchange.

($\implies$): Let $\{Y_t\}$ be a causal ARMA(p, q). Then

$$\text{Causal: } Y_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j} \text{ for some } \{\psi_j\} \text{ such that } \sum_{j=0}^{\infty} |\psi_j| < \infty,$$

$$\text{ARMA}(p, q) : \Theta(B)\epsilon_t = \phi(B)Y_t = \phi(B)\Psi(B)\epsilon_t \text{ where } \Psi(z) = \sum_{j=0}^{\infty} \psi_j z^j \text{ (since causal).}$$

Since $\phi(z)$ and $\Psi(z)$ are power series, $\phi(z)\Psi(z)$ is too; call it $\eta(z)$. Since $\sum |\psi_j| < \infty$, we have $\sum |\psi_j||z|^j \leq \sum |\psi_j|$ on $|z| \leq 1$. So, by the Weierstrass M -test, the radius of convergence is at least 1, i.e., for all $z \in \mathbb{C}$ with $|z| \leq 1$ we have absolute and uniform convergence of $\sum \psi_j z^j$. So, for $\eta(z)$ we focus on the common domain of analyticity: $\mathbb{C}$ for ϕ , intersected with $\mathbb{D}$ for Ψ .

Let

$$\eta(z) = \sum_{j=0}^{\infty} \eta_j z^j, \quad |z| \leq 1. \quad (4)$$

We look at the following relation again and compare:

$$\Theta(B)\epsilon_t = \eta(B)\epsilon_t \Rightarrow \sum_{j=0}^q \theta_j \epsilon_{t-j} = \sum_{j=0}^{\infty} \eta_j \epsilon_{t-j}.$$

Then

$$\text{Cov} \left(\sum_0^q \theta_j \epsilon_{t-j}, \epsilon_{t-k} \right) = \text{Cov} \left(\sum_0^\infty \eta_j \epsilon_{t-j}, \epsilon_{t-k} \right) \quad \forall k = 0, 1, 2, \dots$$

This is simply an inner product: since $\{\epsilon_s\}_s$ has mean 0 and is also uncorrelated, only $j = k$ will survive.

- $\theta_k \text{Cov}(\epsilon_{t-k}, \epsilon_{t-k}) = \eta_k \text{Cov}(\epsilon_{t-k}, \epsilon_{t-k})$ for $k = 0, \dots, q \iff \theta_k \sigma^2 = \eta_k \sigma^2 \iff \theta_k = \eta_k$.
- $0 = \eta_k \text{Cov}(\epsilon_{t-k}, \epsilon_{t-k}) = \eta_k \sigma^2 \Rightarrow \eta_k = 0$ for $k > q$.

Now, since the $\{\eta_j\}$ are known, we have the exact structure of $\eta(z)$ from (4):

$$\eta(z) = \sum_{j=0}^\infty \eta_j z^j = \sum_{j=0}^q \theta_j z^j = \Theta(z), \quad |z| \leq 1 \quad \Rightarrow \quad \phi(z)\Psi(z) = \Theta(z), \quad |z| \leq 1.$$

Since $|\Psi(z)| < \infty$ on $|z| \leq 1$, if $\phi(z_0) = 0$ for some $z_0 \in \mathbb{D}$, then $\phi(z_0)\Psi(z_0) = 0$, and then $\Theta(z_0) = 0$ too. So ϕ and Θ would have the common zero z_0 ! The only way out is that, in $|z| \leq 1$, ϕ has no zeros.

($\Leftarrow$): Since ϕ has no zero in $|z| \leq 1$ and ϕ is a polynomial, $1/\phi$ is an analytic function on a disc bigger than $\mathbb{D}$ (at least slightly bigger: $1/\phi$ is analytic wherever ϕ has no zero, and ϕ has only finitely many zeros; hence we can pick $\varepsilon > 0$ such that $1 + \varepsilon < \min |z_j|$, where the z_j are the zeros of ϕ). We look at the power series expansion of $1/\phi$ in $|z| < 1 + \varepsilon$ for some $\varepsilon > 0$:

$$\frac{1}{\phi(z)} = \sum_{j=0}^\infty \xi_j z^j =: \xi(z). \quad (1)$$

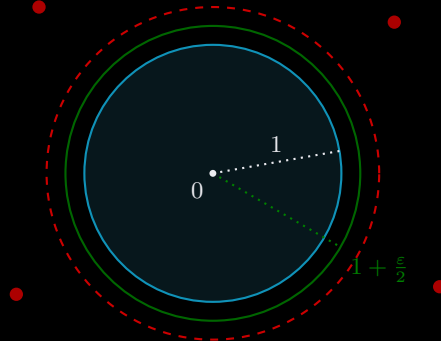


Figure 1.8: The unit circle (solid), the circle of radius $1 + \varepsilon/2$, and the dashed circle of radius $1 + \varepsilon$; the zeros of ϕ (red dots) all lie outside the dashed circle.

For the next bit of the proof, our goal is to show that $\{\xi_j\}_j$ is absolutely summable, and for that we use the convergence of the power series $\xi(z)$ at a point outside the unit disc; for that we need ϕ to have no root in $|z| \leq 1$ instead of only in $|z| < 1$.

At $z = 1 + \frac{\varepsilon}{2}$ in (1):

$$\xi\left(1 + \frac{\varepsilon}{2}\right) = \sum_{j=0}^\infty \xi_j \left(1 + \frac{\varepsilon}{2}\right)^j < \infty \Rightarrow \xi_j \left(1 + \frac{\varepsilon}{2}\right)^j \rightarrow 0 \text{ as } j \rightarrow \infty.$$

So there exists $K \in (0, \infty)$ such that $|\xi_j| \left|1 + \frac{\varepsilon}{2}\right|^j < K$ for all $j \in \mathbb{N}_0$. (If not, then for any $K \in (0, \infty)$ you can pick $j_K \in \mathbb{N}_0$ such that $|\xi_{j_K}| \left|1 + \frac{\varepsilon}{2}\right|^{j_K} > K$, so $\xi_j \left(1 + \frac{\varepsilon}{2}\right)^j \not\rightarrow 0$.)

In particular,

$$|\xi_j| < \frac{K}{\left(1 + \frac{\varepsilon}{2}\right)^j} \Rightarrow \sum_{j=0}^\infty |\xi_j| < K \sum_{j=0}^\infty \left(\frac{1}{1 + \varepsilon/2}\right)^j = K \cdot \frac{1}{1 - \frac{1}{1 + \varepsilon/2}} < \infty \Rightarrow \sum_{j=0}^\infty |\xi_j| < \infty. \quad (2)$$

Also from (1):

$$\frac{1}{\phi(z)} = \xi(z) \Rightarrow \phi(z)\xi(z) = 1 \quad \text{on } |z| < 1 + \varepsilon. \quad (3)$$

We have

$$\phi(B)Y_t = \Theta(B)\epsilon_t \Rightarrow \underbrace{\xi(B)\phi(B)Y_t}_{\text{from (3)}} = \xi(B)\Theta(B)\epsilon_t \Rightarrow Y_t = \xi(B)\Theta(B)\epsilon_t.$$

Y_t is ARMA(p, q), hence stationary by definition. Since $\phi(B)$ is a linear filter, it gives a stationary time series $\phi(B)Y_t := V_t$, and since $\xi(B)$ is a linear filter again (due to (2)), we have that $\xi(B)V_t$ is stationary again. Basically, $\xi(B)\phi(B)Y_t$ makes sense, and since as functions $\phi(z)\xi(z) = 1$, we have $\phi(B)\xi(B) = \text{Id}$ too.

As functions $\phi(z)\xi(z) = \xi(z)\phi(z)$, but as operators $\phi(B)\xi(B)$ need not equal $\xi(B)\phi(B)$; however, since they are (finite or infinite) polynomials in the operator B being multiplied, we have commutativity, and this can be verified by trivial calculations.

We have

$$Y_t = \xi(B)\Theta(B)\epsilon_t = \frac{\Theta(B)}{\phi(B)}\epsilon_t.$$

Lastly, consider the power series expansion of the rational function

$$\frac{\Theta(z)}{\phi(z)} = \sum_{j=0}^{\infty} \psi_j z^j, \quad |z| \leq 1.$$

Then

$$Y_t = \frac{\Theta(B)}{\phi(B)}\epsilon_t = \frac{\Theta}{\phi}(B)\epsilon_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}.$$

This finishes the other direction. We now simply do the small proof to show that $Y_t = \Psi(B)\epsilon_t$ is a solution to $\phi(B)Y_t = \Theta(B)\epsilon_t$:

$$\phi(B)Y_t = \phi(B)\Psi(B)\epsilon_t = \Theta(B)\epsilon_t.$$

We used $\phi(B)\Psi(B) = \Theta(B)$, since we have this equality for the power series $\Psi(z) = \frac{\Theta(z)}{\phi(z)}$ on $\mathbb{D}$. □

Remark 1.49. *In practice, we will often use the version that “ARMA(p, q) is causal if and only if $\phi(1/z)$ has all roots inside $|z| < 1$ ”, and other relations between roots and coefficients.*

Example 1.50 (Application). **AR(1) process:** $\phi(z) = 1 - \phi_1 z$, $\Theta(z) = 1$. Clearly, ϕ and Θ have no common zeros. By the last theorem, AR(1) is causal if and only if ϕ has no zero in $|z| \leq 1$. We see that the zero of ϕ is at ϕ_1^{-1} , and we need $|\phi_1^{-1}| > 1 \iff |\phi_1| < 1$. So

$$\text{AR(1) is causal} \iff |\phi_1| < 1,$$

or, equivalently, AR(1) has an MA(∞) representation if and only if $|\phi_1| < 1$.

AR(2) process: $Y_t = \epsilon_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2}$, with $\phi(z) = 1 - \phi_1 z - \phi_2 z^2$ and $\Theta(z) = 1$. Again, ϕ and Θ trivially have no common zeros. AR(2) is causal if and only if ϕ has no zeros in $|z| \leq 1$. The roots of $\phi(z)$ are

$$\left(-\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2} \right) \frac{1}{2\phi_2}.$$

..... **Calculation!!**

Definition 1.51 (Invertibility). An ARMA(p, q) process defined as $\phi(B)Y_t = \Theta(B)\epsilon_t$ is called *invertible* if there exists a sequence of constants $\{\pi_j\}_j$ with $\sum |\pi_j| < \infty$ such that

$$\epsilon_t = \sum_{j=0}^{\infty} \pi_j Y_{t-j}, \quad t \in \mathbb{Z}.$$

This is simply the other version of what we called causality: the t^{th} noise ϵ_t is a linear combination of all past data $\{Y_s : s \in \mathbb{Z}, s \leq t\}$. Exactly like before, we have a characterisation theorem.

Theorem 1.52 (Characterising invertible ARMA(p, q)). *Let $\{Y_t\}_t$ be an ARMA(p, q) process of the form $\phi(B)Y_t = \Theta(B)\epsilon_t$ for $\{\epsilon_t\} \sim \text{WN}(0, \sigma^2)$. Suppose the polynomials $\phi(\cdot)$ and $\Theta(\cdot)$ have no common zeros. Then*

$$\{Y_t\} \text{ is an invertible ARMA}(p, q) \iff \Theta \text{ has no zeros in the unit disc } \mathbb{D} \subseteq \mathbb{C}.$$

The coefficients $\{\pi_j\}$ in the expansion $\epsilon_t = \sum_{j=0}^{\infty} \pi_j Y_{t-j}$ can be determined by

$$\pi(z) = \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\Theta(z)}, \quad z \in \mathbb{D}.$$

Moreover, $\{\epsilon_t\}$ defined as above is the unique stationary (and invertible) solution time series of the ARMA(p, q) process of the form $\phi(B)Y_t = \Theta(B)\epsilon_t$.

Question 1.53. What changed from the previous set-up?

For causality we have all past shocks observed, but no observation of the process. It is like keeping a count of the number of balls we are adding or removing at every step, instead of keeping a count of the number of balls collected in the box below. In a causal ARMA(p, q), given ϵ , we try to find Y . Here, given the observations Y , we are trying to figure out the shocks ϵ .

Question 1.54. If $\{\epsilon_t\} \sim \text{WN}(0, \sigma^2)$ is already known, how is “finding out” $\{\epsilon_t\}$ of any use to us?

There can be many processes in the class $\text{WN}(0, \sigma^2)$ that satisfy $\phi(B)Y_t = \Theta(B)\epsilon_t$. We pinpointed the unique process ϵ that will give rise to the observed Y .

Theorem 1.55. *If $\{Y_t\}$ is a stationary solution of the equation $\phi(B)Y_t = \Theta(B)\epsilon_t$, where $\{\epsilon_t\} \sim \text{WN}(0, \sigma^2)$, and if $\phi(z)\Theta(z) \neq 0$ on $\mathbb{D}$, then $\{Y_t\}$ is the unique causal and invertible solution, i.e.,*

$$\bullet Y_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j} \text{ with } \Psi(z) = \sum_{j=0}^{\infty} \psi_j z^j = \frac{\Theta(z)}{\phi(z)}, \quad |z| \leq 1,$$

and

$$\bullet \epsilon_t = \sum_{j=0}^{\infty} \pi_j Y_{t-j} \text{ with } \pi(z) = \sum_{j=0}^{\infty} \pi_j z^j = \frac{\phi(z)}{\Theta(z)}, \quad |z| \leq 1.$$

In modelling, we will always take causal and invertible ARMA(p, q) processes.